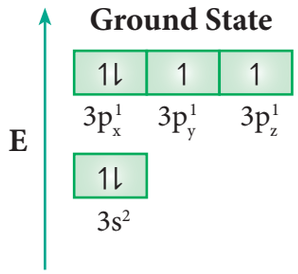
sp^3d^2 Hybridisation:

In sulphur hexafluoride (SF_6) the central atom sulphur extend its octet to undergo sp^3d^2 hybridisation to generate six sp^3d^2 hybridised orbitals which accounts for six equivalent S-F bonds. The ground state electronic configuration of sulphur is $[Ne]3s^2\ 3p_x^2\ 3p_y^1\ 3p_z^1$.



One electron each from 3s orbital and 3p orbital of sulphur is promoted to its two vacant 3d orbitals (d_{z^2} and $d_{x^2-y^2}$) in the excite state. A total of six valence orbitals from sulphur (one 3s orbital, three 3p orbitals and two 3d orbitals) mixes to give six equivalent sp^3d^2 hybridised orbitals. The orbital geometry is octahedral as shown in the figure.

Overlap with 2pz orbitals of fluorine:

The six sp^3d^2 hybridised orbitals of sulphur overlaps linearly with $2p_z$ orbitals of six fluorine atoms to form the six S-F bonds in the sulphur hexafluoride molecule.

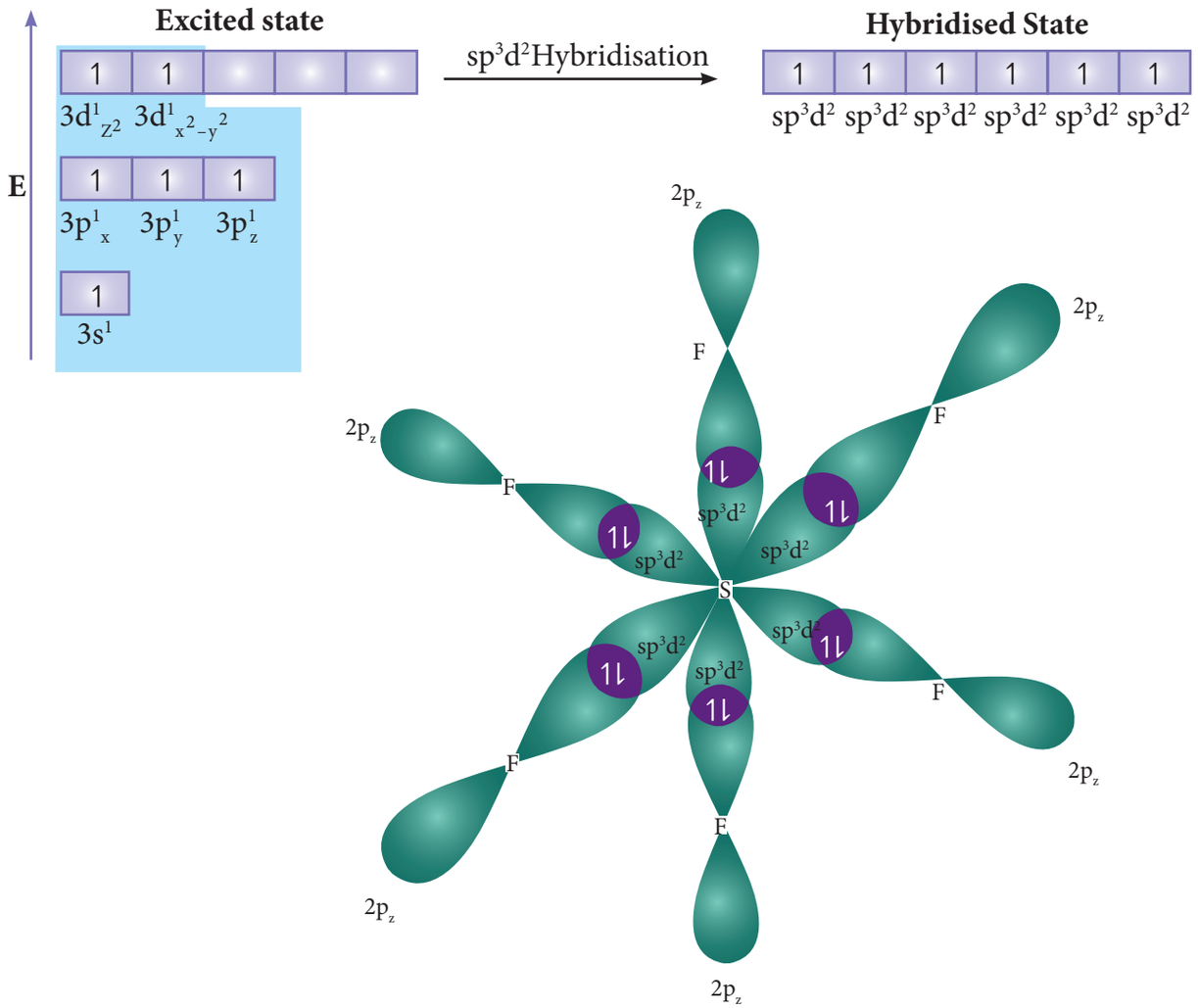


Fig 10.26 sp^3d^2 Hybridisation : SF_6

Unit 10 layout - jagan.indd 94 22-01-2021 12:33:27